Answer Key - Health Sciences - Graduate Level Assessment 23 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Alone; B) Driving during peak traffic hours; C) In a parking lot; D) On a highway; E) Driving at night
   Note: Older adults are most likely to be involved in a traffic accident when they are alone because they may lack immediate assistance or support to navigate difficult driving situations, increasing the likelihood of errors or accidents.

2. **Answer:** E
   Options: A) 2-3 minutes.; B) 30-50 seconds.; C) 0-10 seconds.; D) 1-2 minutes.; E) 10-30 seconds.
   Note: The correct answer of 10-30 seconds is supported by research indicating that holding a stretch within this duration effectively enhances flexibility and muscle elasticity while minimizing the risk of injury.

3. **Answer:** A
   Options: A) Buccinator; B) Orbicularis oris; C) Trapezius; D) Frontalis; E) Sternocleidomastoid
   Note: The buccinator assists in the elevation of the mandible by helping to keep food positioned between the teeth during chewing, thereby facilitating the initial movement of the mandible.

4. **Answer:** A
   Options: A) Extraversion; B) Primary control; C) Narcissism; D) Openness to experience; E) Conscientiousness
   Note: High levels of self-care behaviors are often linked to extraversion because extraverted individuals typically exhibit greater social engagement and proactive health-related behaviors, contributing to their overall well-being.

5. **Answer:** D
   Options: A) converted into citrulline and released from the muscle.; B) converted into urea and released from the muscle.; C) used to synthesise purines and pyrimidines in the muscle.; D) converted into alanine and glutamine and released from the muscle.; E) converted into glycine and released from the muscle.
   Note: The correct answer is based on the fact that during the deamination of branched chain amino acids in muscle, ammonia is primarily converted into non-toxic forms such as alanine and glutamine to facilitate safe transport and release from the muscle.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: Fast-twitch muscle fibers primarily rely on anaerobic metabolism for energy, resulting in fewer mitochondria and lower ATPase activity compared to slow-twitch muscle fibers, which are adapted for endurance and aerobic metabolism.

7. **Answer:** False
   Options: A) True; B) False
   Note: A normal resting heart rate for an adult is generally considered to be between 60 and 100 beats per minute, making the statement false as it inaccurately restricts the range.

8. **Answer:** True
   Options: A) True; B) False
   Note: Huntington disease exhibits anticipation because the severity and onset of symptoms tend to increase with successive generations, particularly when the mutated gene is transmitted from father to child, due to the unstable expansion of CAG repeats in the HTT gene.

9. **Answer:** False
   Options: A) True; B) False
   Note: Eosinophilic oesophagitis is a chronic inflammatory condition that often requires medical treatment, such as corticosteroids or dietary management, in addition to lifestyle modifications to effectively control symptoms and inflammation.

10. **Answer:** True
    Options: A) True; B) False
    Note: The disenchantment phase is characterized by feelings of disappointment and a lack of purpose after the initial excitement of retirement, which often leads individuals to seek re-engagement in work or activities, making it a less common experience compared to other phases of retirement transition.

Long Form Response

11. **Answer:** Proprioceptive information plays a crucial role in the functioning of the temporomandibular joint (TMJ) by providing the central nervous system with feedback regarding the position and movement of the jaw. This sensory information is primarily carried by the masseteric and auriculotemporal nerves, which innervate the muscles and surrounding structures of the TMJ. The masseteric nerve, a branch of the mandibular nerve, transmits proprioceptive signals from the masseter muscle, while the auriculotemporal nerve conveys sensory information from the temporomandibular joint itself. This proprioceptive feedback is essential for the coordination of jaw movements, facilitating functions such as chewing and speaking, and ensuring the stability and integrity of the joint during dynamic activities. Understanding the contributions

of these nerves enhances the knowledge of TMJ disorders and informs therapeutic approaches in dental and physical health.
Note: correct because it accurately describes how proprioceptive information from the masseteric and auriculotemporal nerves informs the central nervous system about the position and movement of the jaw, which is essential for the proper functioning of the temporomandibular joint.

12. **Answer:** The classic glycogen loading regime, which involves depleting glycogen stores followed by a high-carbohydrate intake, may not enhance performance during prolonged endurance exercise due to several factors. Firstly, the process of glycogen depletion can lead to increased muscle fatigue and affect overall training quality in the days leading up to an event. Additionally, recent research suggests that a more moderate carbohydrate intake, maintained consistently, may promote better glycogen utilization and energy availability without the adverse effects of depletion. Furthermore, individual variability in metabolism and the type of endurance exercise being performed can also influence the effectiveness of glycogen loading, rendering it less universally applicable than once thought. Thus, while the classic regime was once widely endorsed, contemporary strategies focusing on a balanced and sustained carbohydrate intake may prove more effective for optimizing performance.
Note: correct because it identifies that glycogen depletion can cause muscle fatigue and negatively impact training, while also highlighting that consistent moderate carbohydrate intake may be more effective for optimizing glycogen utilization during prolonged endurance exercise.

End of Answer Key